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The Central Governor Model of Exercise Regulation Applied to the Marathon

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Abstract

Two popular models hold that performance during exercise is limited by chemical factors acting either in the exercising muscles or in the brain producing either ‘peripheral’ or ‘central’ fatigue, respectively. A common feature of both models is that neither allows humans to ‘anticipate’ what will happen in the future and modify their exercise response accordingly. The peripheral fatigue model predicts that exercise terminates only after there has been catastrophic failure in one or more body systems and only when all the available motor units in the active muscles have been activated. The marathon race provides evidence that human athletes race ‘in anticipation’ by setting a variable pace at the start, dependent in part on the environmental conditions and the expected difficulty of the course, with the capacity to increase that pace near the finish. Marathoners also finish such races without evidence for a catastrophic failure of homeostasis characterised by the development of a state of absolute fatigue in which all the available motor units in their active muscles are recruited. These findings are best explained by the action of a central (brain) neural control that regulates performance in the marathon ‘in anticipation’ specifically to prevent biological harm.

A popular model holds that performance during exercise is limited by chemical factors acting in the exercising muscles producing what is termed “peripheral fatigue.”[1] Thus, during exercise of maximal intensity, it is believed that the heart is unable to supply oxygen to the muscles at a rate sufficient to prevent the development of skeletal muscle anaerobiosis[2] accelerating the production of the toxic metabolite lactic acid, which accumulates and ‘poisons’ muscle function causing fatigue. During more prolonged exercise, glycogen-depleted muscles cannot generate sufficient adenosine triphosphate (ATP), so exercise intensity falls until the rate of ATP production from alternate fuels, especially fats, insures that the muscles cannot become ATP deficient.

This model predicts that homeostasis, classically defined by Cannon[3] (page 24) as a state of relative constancy (equilibria) in the coordinated physiological processes within the living organism, fails catastrophically in the exercising muscles at exhaustion thereby ‘limiting’ human exercise performance. A variant of this model, the central fatigue model, holds that the same homeostatic failure occurs in the brain with either the accumulation of ‘toxic’ concentrations of neurotransmitters or the depletion of other critical neurochemicals.[4,5]

These models predict that humans have no capacity to act ‘in anticipation’ of what might occur in the future. Rather, humans can only exercise mindlessly, as automatons, until a catastrophic metabolic failure occurs either in muscle or brain. The periph-

Table I. Physiological predictions of the peripheral fatigue model that have yet to be shown to be true[6]

Exercise	Current evidence
High intensity exercise	
cardiac output reaches a plateau during maximal exercise	All except two studies disprove this
skeletal muscle anaerobiosis develops during maximal exercise	Disproven
lactic acidosis due to anaerobiosis inhibits skeletal muscle function	Disproven
all motor units are always recruited at the point of exhaustion	Disproven
loss of homeostasis at exhaustion	Disproven
fatigue is purely a physical event	The theory that fatigue is purely an emotion has been proposed[7]
Prolonged submaximal exercise	
exercise terminates when muscles glycogen content equals zero	There is always some glycogen present at exhaustion
exercise terminates when muscles become energy depleted	No evidence that muscles are energy depleted at exhaustion[8]

eral fatigue model also predicts that only factors that alter skeletal muscle metabolism can influence human exercise performance. However, a number of physiological findings do not support this model (table I).[6]

There is no evidence that homeostatic control fails at exhaustion in any form of exercise.[6] For example, with the exception of marked reductions in the partial pressure of oxygen in arterial blood (PaO_2) and partial pressure of carbon dioxide in arterial blood, the physiological disruption during maximal exercise at extreme altitude is less than during maximal exercise at sea level. In particular, heart rate and cardiac output are submaximal as is the extent of skeletal muscle recruitment; blood lactate concentrations are low and there is no metabolic acidosis. Yet, the physical incapacity is extreme. Hence, exercise at extreme altitude cannot be 'limited' by the usual variables (higher heart rate, elevated cardiac output, maximal blood lactate concentrations or severe metabolic acidosis) considered to produce 'peripheral fatigue' at sea level. Rather, exercise is likely limited by the extremely low PaO_2 presumably acting in the brain.[2,6]

Similarly, prolonged exercise terminates without evidence that mitochondrial ATP production is limited by an inadequate supply of metabolic precursors including glycogen.[8] If an inability to rapidly generate sufficient ATP is an exclusive cause of 'fatigue' during prolonged exercise, then muscle rigor would always develop.[2] But, this does not occur.

In contrast, the single variable that is always maximal at exhaustion during all forms of exercise is the rating of perceived exertion (RPE).[9] This phenomenon cannot be explained by the peripheral fatigue model.

The peripheral model also predicts that exercise will terminate only after all the motor units in the active muscles have been recruited.[6,7,10] There is no known mechanism by which 'poisonous' metabolites can directly inhibit the function of skeletal muscle fibres that are not contracting since they have yet to be recruited by the brain. However, metabolites might act via sensory feedback mechanisms acting either in the brain or in the spinal cord to regulate the extent of skeletal muscle recruitment (according to the Central Governor model[7,10,11]).

Thus, the peripheral model predicts that exercise continues only for as long as there are fresh, unrecruited muscle fibres that can be activated to supplement the function of already recruited fibres that are fatiguing.[6,7,10] Finally, when all exercising muscle fibres have been recruited and fatigued, exercise must either terminate or reduce in intensity. However, fatigue can develop before all the muscle fibres in the exercising limbs have been recruited.[6] Thus, we have argued that the peripheral model needs to be replaced.[6]

The Central Governor model[7,10,11] predicts that neural control systems in the brain and spinal cord establish the number of motor units that are activated in the exercising muscles, specifically to ensure that homeostasis is maintained regardless of the exercise intensity or duration, the environmental conditions in which the exercise is performed[12] or the athlete's biological state. Sensory information from the periphery is integrated by the brain to determine appropriate exercise behaviours that ensure bodily homeostasis. This model adds to those which focus on the role of sensory feedback from muscle and elsewhere in the regulation of cardiovascular function during exercise[13] since it considers how such sensory feedback might also modify the exercise performance (as opposed to cardiovascular function alone). Furthermore, this Central Governor model predicts that the brain also generates the perception of fatigue (the RPE) specifically to ensure that rising levels of discomfort cause the termination of exercise before homeostasis fails.[9,12]

In summary, the novel prediction of this model is that each exercise performance is regulated by the recruitment of the exactly appropriate numbers of muscle fibres to ensure that the exercise is completed safely.

1. How Do the Peripheral and Central Governor Models Explain Some Common Phenomena Experienced by Marathon and Other Runners?

Athletes run races of different distances at different paces, and the paces are established early in exercise, perhaps within the first few steps. The peripheral model predicts that when 'poisonous' metabolites in the active muscles reach their toxic concentration, they impair the function of the contracting skeletal muscle fibres, thereby establishing the pace. In the words of Pfitzinger and Douglas,[14] "…your marathon pace is very close to your lactate threshold pace, which is determined by your oxygen consumption at your lactate threshold and your running economy. If you run faster than your lactate threshold pace, then lactate accumulates in your muscles and blood; this occurrence deactivates the enzymes for energy production and makes you slow down."

This explanation cannot be correct because it cannot explain why athletes begin exercise of different durations at different paces. Rather, this model must predict that athletes will always begin exercise at a very fast pace and then slow down once the poisonous metabolites achieve their 'toxic' levels. Second, this model predicts that humans must run at one pace regardless of the distance they wish to cover. This pace will be the speed at which "lactate accumulates in your muscles and blood…and makes you slow down."

However, athletes clearly chose their pace 'in anticipation' of the expected duration of the upcoming event. For example, the pace chosen in world record performances at distances from 800 to 10 000m[15] is different, even from the start of exercise, and is slower the longer the race distance. Only the Central Governor model can explain this because it allows the initial pace to be set by the brain 'in anticipation' of the expected duration of the exercise bout. This it achieves by recruiting either more (in races of shorter distance) or fewer (in races of longer distance) fibres (motor units) in the active skeletal muscles. Furthermore, this model allows lactate to act as an essential metabolite with multiple functions,[16] rather than as a simple 'poison'. Indeed, the foundation concept of the peripheral model (the ability of lactic acid to act as a peripheral regulator or 'governor' of exercise performance) is no longer true.[16]

Athletes are able to increase their pace near the end of races lasting more than a few minutes (the end spurt). Thus, the analysis of all world record performances in track athletics shows that, with the exception of the 800m and to a lesser extent the mile, athletes speed up at the end of the longer races.[15] Similarly, the final 100–400m of a close finish in the marathon is often the fastest section of the race. But, the peripheral model cannot explain how all the recruited and already fatigued muscle fibres can each suddenly increase its power output simply because the race is nearly finished. How, for example, do these exhausted fibres sense that the race is about to finish, unless so informed by the eyes and the brain?

The peripheral model predicts that fatigue is absolute and a period of rest must be allowed before any further activity can be undertaken.[7] But, even the most exhausted marathon runner can still walk before or after finishing the race and so is not totally exhausted. Nor can the peripheral model explain why fatigue appears to be an emotion,[7] which increases as a function of the duration of exercise that remains and which is established almost from the onset of exercise.[9]

The peripheral model cannot explain how factors that do not alter skeletal muscle metabolism can influence exercise performance. Interventions that act exclusively by altering brain function and which may also alter exercise performance include hypnosis, music, psychological training and certain performance-enhancing drugs such as the amphetamines. Especially the actions of the amphetamines, which seem to impair the regulatory effects of the Central Governor model, support the theory that the brain is the crucial determinant of human exercise performance.

The presumption is that training improves performance principally by increasing the capacity of the trained skeletal muscles to generate more energy, more rapidly. It seems improbable that the brain remains unchanged by the training process. Perhaps the statement made in 1987 by the German sports physician Heinz Liesen that "every adaptation to training occurs in the brain and nervous system" is a slight exaggeration.

2. Summary

The Central Governor model predicts that performance during the marathon is set by the subconscious brain specifically to ensure that the athlete reaches the finish line whilst still in physiological homeostasis. The model also predicts that each athlete finishes every race with some physiological reserve and could therefore have run slightly faster. Whilst this may be of little interest to race winners, it does raise novel questions of the possible biological factors that determine which athlete finishes second.

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